A New Explainable Deep Learning Framework for Cyber Threat Discovery in Industrial IoT Networks

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***Abstract*—Industrial Internet of Things (IIoT) and Industry 4.0 empower interrelation among manufacturing processes, industrial machines, and utility services. The time-critical data collected from heterogeneous sensing devices are usually communicated to processing points for analysis and aggregation as the basis of IIoT. The IIoTs' service quality typically depends on data integrity and accuracy, which could be exploited by injecting malicious events, such as false data injection and data poisoning attacks. Thus, effective anomaly recognition and explanation are critical for ensuring quality services and empowering security administrators to interpret the causal reasoning of prediction decisions and underlying data evidence. This study proposes an autoencoder-based detection framework using convolutional and recurrent networks to discover cyber threats in IIoT networks and explain the model. A two-step sliding window (SW) is applied to learn the latent representations of data features better. Malicious points from the raw time series are transformed into fixed-length series through the first-step SW. Every series is converted into continuous-time-reliant subseries via another smaller SW to learn latent representations of malicious events. Fully connected networks use the extracted temporal and spatial features for the classification and explanation of attack events. The empirical results revealed that this framework effectively extracts features that include contexts of malicious patterns. This demonstrated that the proposed framework is robust in detecting malicious events using multiple evaluation metrics and outperforming the contemporary state-of-the-art methods, indicating its suitability as an operative application method in real-world IIoT-based networks.**

***Index Terms*—Anomaly detection, Industrial Internet of Things (IIoT), industrial networks, time series.**

I. Introduction

INDUSTRIAL Internet of Things (IIoT) systems are progressively incorporated in cyber–physical systems (CPSs), comprising critical infrastructures (CIs), for example, smart grids, utility plants, gas pipelines (GSPs), water management and treatment systems, and many more. In such networks, IIoT systems are usually part of an industrial control system (ICS) or Internet ICS (IICS), expected with the enduring and reliable functioning of CI networks. IICSs can be generally indicated to contain MODBUS protocols, programmable logic controllers (PLCs), and supervisory control and data acquisition (SCADA) systems.

The linking of IIoT with IICS networks exposes them to different vulnerabilities and cyberattacks that disrupt the operations of these CIs. Recently, in 2018, several successful attacks on GSP systems were reported in the U.S, which ended in the stoppage of utility services and customer communication networks for many days [1]. One of the prominent examples is the Stuxnet attack, which compromised nuclear plants, prompting severe damage to the CIs [1], [2]. Another example is the incident of a water plant that targeted to disturb the functioning of a pump which eventually caused the failure in plant [3]. Similarly, the power grids were attacked in 2015 by an attack named *BlackEnergy3*, causing the outage of power to around 230 000 persons [4]. Even though security mechanisms implemented for operational technology (OT) and information technology (IT) systems are reasonably advanced, they could not be openly employed by IICSs because of the close-fitting between cybersystems and the controlled physical processes.

Hence, security mechanisms that are based on the system level are essential to examine the behaviors of physical processes and continued operational availability of the system [1]. Unlike most OT/IT systems, the security aims of IIoT-based IICSs are ranked in availability, integrity, and confidentiality order [5]. Due to the close connection between physical processes and feedback control loop variables, effective attacks on IICSs can cause serious damage to CIs and possibly fatal outcomes for a country's humans, environment, society, and economy. This highlights the importance of scheming a robust and efficient security solution that can detect cyberattacks in real time and avoid intrusions pointing toward IICSs networks [1].

Current cyberattack discovery models are generally based on two kinds: 1) anomaly and 2) signature. To alleviate the recognized drawbacks in both anomaly-based and signature-based intrusion detection methods, several attempts have been made to develop strategies that are hybrid [6], [7]. Even though the methods based on hybrid approaches are proficient at identifying malicious activities, they are not trustworthy due

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to recurrent updating in network strictures, causing diverse intrusion detection system (IDS) taxonomies. Apart from this, the conventional attack recognition approaches mostly depend on the analysis of network metadata, i.e., duration of traffic, intervals of packets, ports of transmission and IP addresses. Hence, there has been repeated focus in applying attack discovery solutions built on machine learning (ML) or deep learning (DL) techniques in recent times. The attack recognition methods can be characterized as host- or network-based techniques. The *K*-nearest neighbor (KNN), decision trees, naive Bayes, and deep neural networks (DNNs) are generally applied approaches for detecting attacks using network traffic flow. These methods examine network traffic data in real time to identify abnormal activities in a timely way. Nevertheless, attack discovery often fails to identify insider or sophisticated attacks.

Unsupervised learning-based security models, especially IDS, monitor data related to physical systems as they do not rely on comprehensive information of the attacks. In common, a knowledgeable attacker with adequate time and information, i.e., an advanced persistent threat actor in the nation-state, can likely evade robust security mechanisms. Moreover, most of the contemporary techniques overlook the problem of imbalanced data by forming only a normal behavior profile of a system and distinguishing deviances from that normal systems' profile as abnormalities. This could be because existing data sets lack real-world setups and limited attack data. Even though consuming the observations of the majority class is a viable method to circumvent problems due to imbalanced data sets, the learning network will have no interpretation of the patterns of attacked data.

An IDS suffers from a high rate of false positives and fails to identify unseen, or zero-day attacks [8]. Therefore, numerous attempts have been made by researchers to employ DL-based models, e.g., to assist the learning from the automated representation of features to build simpler models from complex ones [9] without relying on features that require massive human effort [10]. The abnormal data produced by particular events in IIoT-based networks mean that situation of the observance field has evolved considerably; thus, the alarm signals need to be activated and handled instantaneously. The abnormal events of data occurrence triggered by energy exhaustion or hardware (sensor) failure imitate the issues of sensor networks, which require replacing affected nodes. The measurement errors in data could affect the accuracy of IIoT devices and the service quality of IICSs networks, which unavoidably needs to be identified and handled efficiently. The data produced from sensors typically comprise of series of times that have distinctive sequential and temporal characteristics. Thus, unique approaches of anomaly recognition for IIoT-based time series are essential.

Most of the existing intrusion detection approaches are built on traditional ML methods. Recently, various studies have been developed based on DL algorithms. The utmost illustrative models of DL are the CNNs and RNNs. The earlier is mainly employed in processing images, having several kernels at the convolution layer (CL) to mine spatial information and local features. The latter is used in NLP and speech recognition, applying memory units to mine time-based information from sequential data. There are multiple employments of memory units of RNN, of which the most well known are the gated recurrent unit (GRU) [11] and long short-term memory (LSTM). Both GRU and LSTM apply diverse gates to overcome the fading gradient problem produced by the loss of memory for sequences that are long term in nature. As RNN and CNN can effectively mine temporal and spatial features from data, scholars made an effort to pool these two techniques for enhanced prediction and detection in time series [12]. Kim and Cho [13] employed the method of C-LSTM for anomaly detection in time-series data. The C-LSTM method involves CNN, LSTM, and DNN. Initially, the univariate time-series data are divided into several fixed-length sequences utilizing a sliding window (SW) to build the structural data set. Then, high-ranking temporal and spatial features are mined using CNN and LSTM. Finally, the mined features are then forwarded to the DNN for classification purposes. In this study, we propose an enhanced model for efficient intrusion detection based on [13] to detect cyberattacks in IIoT-based IICSs setups and interpret the prediction decisions to enable security administrators to understand the underlying data evidence and causal reasoning.

In this work, another minor SW is utilized to create subsequences that are time-dependent, which makes it more effective for extracting temporal features. It is worth mentioning that a CL with 1-dimension is applied in our proposed method as it is broadly employed in processing sequences and could be deliberated as a 2-dimension CL having the height of one. The extracted features by CNN are then forwarded to an LSTM-based autoencoder with the target to learn features at the reconstruction level to make them more advantageous during attack detection using DNN. The evaluation outcomes using a real-world industrial networks-based data set, i.e., the GSP network [14] verifies that our proposed framework can accomplish a higher detection rate on several performance metrics and less rate of false positives. The main contributions of this work are as follows.

1) A novel explainable Conv-LSTM-based autoencoder framework is proposed for time-series-based attack detection in IIoT networks. The framework is novel in the sense that it integrates and explains the advantages of CNN and LSTM for discovering complicated cyberattacks in IIoT-based networks.
2) The proposed model is capable of detecting both seen and unseen (*zero-day*) attacks from IIoT setups. The experimental results indicated the superiority of the proposed method compared to compelling methods and show that it is robust to imbalanced data.
3) The phenomenon of SWs is utilized for the processing of time-series data. The outcomes indicate that this process is efficient in improving the performance of combined LSTM and CNN models.
4) The XAI concepts are explored to enhance trust management and enable security administrators to interpret the causal reasoning of prediction decisions and underlying data evidence. The analysis of feature importance is studied for improved anomaly detection.

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The remainder of the article is structured as follows. In Section II, related work regarding time-series-based anomaly detection and DL are reviewed. The proposed framework is presented in Section III, while the evaluation results and experimental setup details are discussed in Section IV. Finally, Section V concludes the study and presents some future work.

II. Background and Related Work

This section discusses the studies related to this work and critical approaches employed for time-series data mining and IDSs.

A. Time-Series Data Mining

Esling and Agon [15] proposed a comprehensive survey of time-series data mining (TSDM). It delivers a broad overview and deep comprehension of TSDM. They defined time series (ts) as a group of values acquired from uninterrupted measurements through time. The authors of this study characterized the ts tasks into seven categories: 1) anomaly detection; 2) segmentation; 3) motif discovery; 4) prediction; 5) clustering; 6) classification; and 7) query by content. Even though there are different goals of TSDM tasks, the three most general issues common to mining are indexing, similarity, and representation techniques. Since the time-series data are of high dimensions, they are vital to decrease the dimensions by effective representation techniques while preserving essential information. Specifying the method for representation, the process of distance measurement or similarity measurement between every sequence in ts needs to be accurately determined for enhanced clustering or classification task. The indexing methods are applied in large databases to retrieve and manage big data.

Giannoni *et al.* [16] briefly discussed the implementation of conventional ML models in anomaly discovery. Likewise, Kim and Cho [13] showed that the process of anomaly recognition is a procedure of mining data attributes and recognizing their usual or anomalous behavior layouts. From the viewpoint of extracting features, the authors separated the existing detection studies into three kinds: 1) mining spatial features; 2) mining temporal features; and 3) statistical modeling. In recent times, the current literature only studies for mining temporal or spatial features distinctly. Hence, the studies in [17] and [18] extracted high-level information by combining CNN and LSTM. The rest of this section discusses the related work regarding RNNs and CNNs.

Hubel and Wiesel [19] proposed CNN by studying the visual cortex of monkeys and cats. Their study discovers that the visual cortex contains neurons that individually respond to lesser regions. The receptive field is defined as the specific region of the sensory area in which a stimulus will adjust the neuron firing. This receptive field study delivers the theoretical foundation for the insights of CNN, which comprises three layers: 1) input; 2) hidden; and 3) output layers. The hidden layer part holds CL, a fully connected layer (FCL), an activation layer (AL), and a pooling layer (PL). The core layer of the CNN (i.e., the CL) is motivated by the study of the receptive field. It computes the data convolution from the input layer by utilizing kernels or filters to mine high-level information about the spatial features. Since the features mined by adjacent filters in CL are identical, the PL is responsible for a downsample or reducing the number of features by selectively removing unnecessary features while preserving vital information.

The CNNs' outstanding outcome of extracting spatial features makes scholars employ it to TSDM. Nevertheless, it is commonly applied for image processing having a 2-D convolution kernel. As time series consists of univariate or multivariate sequences, it requires to be converted into distinctive formats. Liu *et al.* [20] proposed a method based on a tensor design that also used CNN for categorizing multivariate time series. Although these approaches effectively mine spatial attributes from the ts and enhance the efficiencies of various algorithms, however, Kim and Cho [13] demonstrated that the loss in data regarding temporal dependencies might happen in CL and PLs when mining spatial attributes from ts.

RNNs are considered as well suited for sequential data processing. Typically, neural networks are linked between the layers using weights. RNN is a special kind of neural network because weights also link the same layer of neurons. A typical RNN structure is composed of input, hidden, and output layers, where the hidden layers' state could variate through time. The hidden layer neurons are connected to both within layer neurons and the input and output layers. The current time t in the hidden state is produced by new input data at time t and the hidden state at the preceding time $t-1$. These hidden units preserve part of the historical information from input data at every time step. Nonetheless, when processing long input sequences, the loss of historical information may occur because of the restricted capacity of hidden units, resulting in a gradient vanishing problem.

Similar to CNN, numerous researchers applied RNN in TSDM for extracting temporal features. For example, Karim *et al.* [21] proposed a method for time-series classification by integrating CNN and LSTM models. Their proposed LSTM-based model enhanced the efficiency of the learning network by leveraging from the attention mechanism to detect the class label. Similarly, Malhotra *et al.* [22] applied LSTM to forecast and detect anomalies in time-series data. The temporal information is extracted from hidden units, and historical data using LSTM then forwarded to a fully connected network for multistep prediction.

Autoencoder is the unsupervised category of neural networks [23]. It can deeply mine hidden attributes from incoming data by reconstructing actual data using these features. This architecture can be employed not only to reduce dimensions but also additionally for prediction and classification. The typical architecture of the autoencoder involves an encoder and a decoder, as shown in Fig. 1(a). Sutskever *et al.* [24] proposed a method for sequential data using an autoencoder-based architecture of the RNN model, as displayed in Fig. 1(b). Their proposed RNN-based encoder extracts and remembers the input data temporal features by receiving the incoming sequence every time. The decoder then reconstructs actual or target series by obtaining the encoded information from hidden units.

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B. XAI in IoT Networks

DL methods are often considered black-box models and suffer from deficiency of interpretability and explainability, such as understanding the causal reasoning and underlying data evidence of the prediction decisions for the behavior of the adopted model. XAI assists in building trust in the developed model by explaining the prediction decisions and behavior of the adopted model. Several researchers tried to explain the prediction decisions and behavior or adopted models, such as Nguyen *et al.* [25] developed an explainable design for the monitoring of customer sentiment in IoT environments. They build their method by combining enterprise data and IoT to model customer sentiment, which helps prioritize the customers in a better way and troubleshoot difficulties. Similarly, Hossain *et al.* [26] and Khan *et al.* [27] proposed explainable DL models for the detection of threats in IoMT networks.

C. Related Work

1) Machine Learning: Zolanvari *et al.* [28] compared several ML algorithms for attack detection in ICS systems, i.e., Naïve Bayes (NB) random forest (RF), support vector machine (SVM), logistic regression (LR), DT, ANN, and KNN using the WST data set. They evaluate their method for detecting several attacks, such as injection attacks, command, and backdoors. The comparative results proposed that the RF achieved the best detection rate with 0.97 recall, and LR is the poorest of all by performing 0.47 recall rate. The authors also discussed that KNN, SVM, and LR misclassified several attack observations as normal data. These methods are fragile to imbalanced data and thus are not appropriate for detecting cyberattacks in ICS networks.

Similarly, Khan *et al.* [7] proposed an instance learning-based KNN method for detecting cyberattacks in ICS networks using GSP data. They utilized oversampling technique to balance the data set and manage to achieved *97%* accuracy rate, 98% precision, 92% recall, and 95% f-score. In another work, Das *et al.* [29] proposed an anomaly detection method based on logical analysis of data (LAD). They proposed a 2-step detection method, where the first step determines the state of the system as stable or unstable, and the second step detects the existence of an attack. After comparative analysis with different methods, such as SVM and DNN, the outcomes concluded that LAD achieved better recall and f-score.

2) Deep Learning: James *et al.* [30] proposed a security method for power systems to identify injection attacks using deep networks. They achieved 91.80% accuracy on average for two data sets. Similarly, Aboelwafa *et al.* [31] proposed a method using autoencoder-architecture to distinguish injection attacks and filter them by utilizing denoising autoencoders. Although, experimental outcomes showed that their proposed method achieved better results as compared to SVM. But, during the training process, they ignored attack data to handle the influence of imbalanced samples. To detect cyberattacks in CPS, Yan *et al.* [32] proposed extreme learning machine (ELM)-based approach. They used only normal data in the training stage to handle the imbalanced data issue, and evaluation outcomes indicate that their method performed better than the SVM-based detection model.

Despite effectual performance in traditional ML- and DL-based approaches, most current techniques suffer from dimensionality curse because of large volumes of data created by IIoT-enabled IICS networks. Hence, a feature engineering technique must be applied to reduce the feature space to reduce computational overhead. Additionally, in a real-world IICs scenario, the issue of an imbalanced data set also needs to be addressed, as the attacked data could not be ignored in the learning process. Also, analysis of time-series data in IIoT-based IICS networks appears to be understudied, and less focus has been paid to the interpretable side of adopted algorithms. Therefore, this study integrates the capability of automatically mining features in the autoencoder architecture intending to take advantage of LSTM in processing sequential data for better feature extraction and attack detection by incorporating data regarding physical processes and networks to overcome the issue of imbalanced data without oversampling or undersampling. In comparison, Kim and Cho's [13] work is based on basic LSTM architecture with a single SW. Additionally, the likelihood of detecting anomalies via reconstruction error is assessed in this research, which is not focused on in Kim and Cho's [13] work. Furthermore, we also evaluate the explainable side of the prediction results developed for anomaly detection in IIoT-enabled industrial networks.

III. Proposed Detection Framework

A. Problem Statement

Let $A = (a^1, a^2, a^3, \ldots, a^m)^T = (a_1, a_2, a_3, \ldots, a_t) \in \mathbb{R}^{m \times t}$ be a time series, where m denotes the number of features and timestamp of every value is denoted by t. The input vector at t time is denoted by $a_t = (a_t^1, a_t^2, a_t^3, \ldots, a_t^m) \in \mathbb{R}^m$. As the input series in some research areas has one feature at every timestamp, so it is considered as univariate series. But, intrusion detection for time series involves two tasks: 1) discovering all the points of irregularities and 2) classifying the target sequences.

The study in [33] focused on finding all the outliers/irregularities while Zhu *et al.* [34] tried to correctly classify/label the target sequence. The prior studies are regression based and the later fits to in the category of classification or clustering. The issue of identifying outliers can be reformed into clustering or classification by employing the concept of SW [35]. The entire series can be fragmented into continuous shorter series by utilizing the SW and 2-D data set $2\text{D} = (2d_1, 2d_2, 2d_3, \ldots, 2d_{t-S+1}) = ((a_1, \ldots, a_S); (a_2, \ldots, a_{S+1}); \ldots; (a_{t-S+1}, \ldots, a_t)) \in \mathbb{R}^{S \times (t-S+1)}$ can be obtained, where the SWs' length is denoted by S and $2d_j$ is deliberated as irregularity if it encompasses any abnormality value from original sequence. Hence, the intrusion detection problem for A is reformed to mark every vector of input for 2-D. This study uses the classification technique to resolve this problem and learning is done through a labeled data set during the training step and

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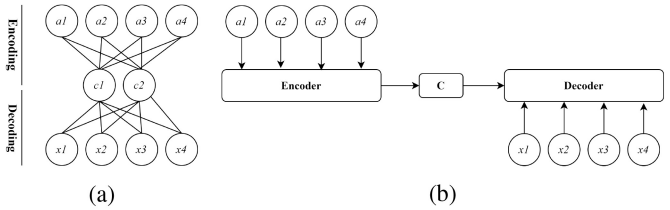


Fig. 1. Autoencoders' architecture. (a) represents the typical autoencoder architecture applied using fully connected neurons while (b) denotes the autoencoder based on RNN architecture which is effective for dealing with sequential information.

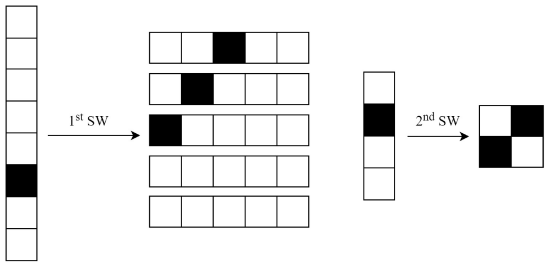


Fig. 2. Architecture of two-step SW.

defined as

$$\max\big(\text{Prob}\big(x_j = 0 \mid 2d_j\big), \text{Prob}\big(x_j = 1 \mid 2d_j\big)\big) \tag{1}$$

whereas $x_j = 1$ denotes the normal sequence and the abnormal sequence is represented by $x_j = 0$ in $2d_j$.

B. Development of Model

Inspired by the work of Kim and Cho [13], the proposed method encompasses CNN and autoencoder-based LSTM architecture, which are utilized to mine high-level temporal and spatial features from ts data. Then, a fully connected neural network is employed to categorize mined features and recognize abnormalities. The abnormal points in the original time series of two data sets are fragmented into numerous series by utilizing the first SW. The sequence is considered anomalous if it contains an abnormal point. Then, as shown in Fig. 2, every series is reformed and restructured into continuous segments by utilizing the second SW. This study considers only univariable time series, and every series could be improved into tensor with one channel and three dimensions. Since the IIoT-enabled IICS networks involve heterogeneous devices, the generated data will be multivariable. This method is well suited, and multivariable ts could be handled as a multichannel tensor with 3-dimensions.

The enhanced CNN model comprises CL with one-dimension, max-PL with one-dimension, and the applied activation function is relu. The spatial attributes at high-level could be mined via CLs, but the adjacent convolution kernels may extract redundant or identical information. Hence, to maintain spatial information, max-PL is employed for the downsampling process. Before the data are given to CL, it is preprocessed by SW, and the 2-D data set is reformed from one-dimension univariable time series. Unlike C-LSTM, a two-step SW is applied to preprocess the ts, where the 1st SW is similar to that in C-LSTM, and the 2nd lower size SW reforms every 2-D series in a data set into fragments that are continuous-time-dependent temporal features. Evaluation results proved that 2-step SW achieved better results using the unified LSTM and CNN model.

The proposed model uses enhanced model where the LSTM architecture is substituted by the autoencoder architecture, motivated from research by Malhotra *et al.* [36] and Ashraf *et al.* [37]. For the autoencoder created on the RNN architecture, the time series $a = (a_1, a_2, a_3, \ldots, a_t) \in \mathbb{R}^{m \times t}$ is given to the encoder, and updating of hds in the encoder is done as

$$\text{hds}_t = f_1(\text{hds}_{t-1}, a_t) \tag{2}$$

where the encoders' hidden state at t is represented by $\text{hds}_t \in \mathbb{R}^n$, the hidden units' size is denoted by n, and the activation function is denoted by f_1.

The fundamental concepts of diverse variants of RNN are similar, but the difference lies in the updating process of the hidden state. The LSTM architecture applies cell and hidden states to secure information about the long short-term dependencies using diverse gates, i.e., forget gate (fg_t), output gate (og_t), and input gate (ig_t) to control cell state (cs_t). The updating is done as

$$fg_t = \sigma\big(WS_{fg}\big[\text{hds}_{t-1}; a_t\big] + bv_{fg}\big) \tag{3}$$

$$ig_t = \sigma\big(WS_{ig}\big[\text{hds}_{t-1}; a_t\big] + bv_{ig}\big) \tag{4}$$

$$og_t = \sigma\big(WS_{og}\big[\text{hds}_{t-1}; a_t\big] + bv_{og}\big) \tag{5}$$

$$cs_t = fg_t \odot cs_{t-1} + ig_t \odot \tanh\big(WS_{cs}\big[\text{hds}_{t-1}; a_t\big] + bv_{cs}\big) \tag{6}$$

$$h\text{ds}_t = og_t \odot \tanh(cs_t) \tag{7}$$

such that the concatenation of the current input is a_t and previous hidden state hds_{t-1} is represented by $[\text{hds}_{t-1}; a_t]$, σ denotes the activation function, and elementwise multiplication is declared by $\odot$.

The weight space (WS) and bias vector (bv) for three gates are the learning parameters, where ig is responsible for governing current input memory flow and fg is responsible for saving information about previous cs. Then, cs and outgoing data will calculate current hds_j. This hds could be deliberated as intermediate data or encoded data which would be utilized in decoding procedures. For the task of prediction, the decoder targets to take knowledge from hds_t to $\tilde{a}_t$ mapping, and attempts to attain the minimum variance between $\tilde{a}_t$ and a_t. For the detection process, the decoder will elicit attributes from hds to recognize enhanced classification outcomes. hds of every t will be given as input to the decoder as new time series, and the decoders' hds'_t will be forwarded as reconstructed features to an FCL for better performance in classification.

As presented in Algorithm 1, the trained model can be applied to identify abnormal time series. Initially, the raw time-series data A is converted into data set 2-D in two dimensions by utilizing the first-step SW, and every portion in the data set, i.e., $2d_j \in 2D$, will be identified by the trained classifier. Every fragment $2d_j$ at this step will generate COS (continuous sequences) by utilizing the second-step

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Algorithm 1 Proposed Detection Framework

Entail:

$A = (a_1, a_2, a_3, \ldots, a_t) \leftarrow$ raw TS data

$S \leftarrow$ the size of 1st step SW

$S' \leftarrow$ the size of 2nd step smaller SW

CNN, autoencoder-based LSTM, and DNN $\leftarrow$ trained models

Ensure:

Class of every fragment in 2D

1: $2D = (2d_1, 2d_2, 2d_3, \ldots, 2d_{t-S+1}) \leftarrow$ Use SW to create 2D dataset
2: **for** each $2D_j$ **do**
3: Create SE_j series from $2d_j$ and size of every series is S'
4: Use CNN to extract spatial features at high-level from SE_j
5: Use autoencoder-based LSTM to extract temporal features at high-level
6: Use DNN for classification using the extracted features
7: Claasify $2d_j$ as malicious if $Prob(x_j = 1 |\ 2d_j) > Prob(x_j = 0 |\ 2d_j)$
8: **end for**

Generate explanations of every detection through LIME

explanation(DP) $=\text{argmin}_{meth}\ LW(fl, meth, \pi_{DP}) + \Omega(meth)$

smaller size SW, and the series can be reflected as temporal attributes at the low level, which will be given to the trained classifier. After this process, the autoencoder based on LSTM and CNN will mine temporal and spatial attributes at a high level from series, which the DNN model will use for classification. Finally, the output of DNN will generate the normal and anomaly probabilities as $\text{Prob}(x_j = 0 \mid 2d_j)$ and $\text{Prob}(x_j = 1 \mid 2d_j)$, respectively. $2d_j$ will be considered as malicious, if $\text{Prob}(x_j = 1 \mid 2d_j) > \text{Prob}(x_j = 0 \mid 2d_j)$; otherwise, it will be categorized as normal series. Finally, the explanation of each prediction decision will be generated using (8) for better understanding interpretation of the results.

C. Explainable AI

We employ local interpretable model-agnostic explanations (LIMEs) to explain the predictions of the proposed Conv-LSTM-based autoencoder framework. Ribeiro *et al.* [38] introduced LIME in 2016 and is also accessible as a python library,[1] which eases the integration of LIME into the development environment. Moreover, LIME has already been explored and applied in several tasks, i.e., facial expressions recognition and estimation of remaining useful life in industrial machinery. After a few experiments, we chose to illustrate the most relevant attributes and their weights as the basis for interpretation in the process of anomaly detection.

The adopted LIME method assesses the incoming data points (DPs) as a function of their proximity to their DPs in the original space. Ultimately, it fits a surrogate model: a linear regression on the DPs having variations through those exemplifying weights. Each DP in the original space can be then understood and clarified with the recently trained prediction algorithm. Precisely, the interpretation or explanation of a DP is a method meth that tries to lessens the locality-aware function of loss $LW(fl, \text{meth}, \pi_{DP})$ measuring how incorrect meth approaches the algorithm to be understood *fl* in its locality *DP* whereas keeping the model complexity to low, and is represented as

$$\exp(DP) = \text{argmin}_{\text{meth}}\, LW(fl, \text{meth}, \pi_{DP}) + \Omega(\text{meth}). \quad (8)$$

We used the LIME model for the XAI threefold.

1) It is widely used and explored in various fields, for example, residual valuable life estimate of industrial machinery and facial expression recognition.
2) The similar surrogate method can be utilized for various AI algorithms.
3) The reliability features of this model offer a better way of interpreting how trustful the explanatory method is in clarifying the black-box detection decisions in the area of attention of data.

TABLE I
EXPERIMENTAL SETUP ENVIRONMENT

System Configuration	Model
Operating system	Ubuntu 18.04.2 LTS
Working environment	Python
Ram	$128Gb$
CPU	Intel(R) Xeon(R)Silver $4110CPU2.10GHz$
Libraries	Keras Scikit-learn TensorFlow Numpy
GPU	NVIDIA $RTX2080GPU$

IV. PERFORMANCE EVALUATION

A. Experimental Setup

The programming language used in the experiments is Python3.7 using Keras and TensorFlow frameworks. The detailed specification of the setup is presented in Table I.

B. Evaluation Data and Evaluation Metrics

A real-world GSP system data source is used in our conducted experiments, created by MSU [14]. This data source contains time-series data with real and synthetic labeled anomaly points. The entire data set consists of 274 628 samples out of which 214 580 are normal data samples, and 60 048 are anomalous. The detailed specification of each record type is listed in Table II.

This data set covers seven broader kinds of cyberattacks, such as Recon, DoS, and Injection attacks. Each sample in this data source is stored in a log file having twenty features and is available as CSV files.[2] Following the method applied in [13], an SW is employed to create continuous fragments where a fragment is reflected as malicious if it encompasses any anomalous point. Through this process, normal and abnormal fragments are created, which constitute the whole data source, and it is distributed into a ratio of 0.2 for training and testing.

[1] https://github.com/marcotcr/lime

[2] https://sites.google.com/a/uah.edu/tommy-morris-uah/ics-data-sets

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TABLE II
ATTACK TYPES OF GSP DATA SET AND THEIR NUMBERS: RECONN = RECONNAISSANCE, MPCI = MALICIOUS PARAMETER COMMAND INJECTION, DOS = DENIAL-OF-SERVICE, MFCI = MALICIOUS FUNCTION CODE INJECTION, NMRI = NAÏVE MALICIOUS RESPONSE INJECTION, MSCI = MALICIOUS STATE COMMAND INJECTION, AND CMRI = COMPLEX MALICIOUS RESPONSE INJECTION

Record Type	Number of Records
Normal	214580
DoS	2176
Reconn	3874
NMRI	7753
MFCI	4898
MSCI	7900
CMRI	13035
MPCI	20412

We treated anomaly detection as a classification problem in our experiments; the proposed method is evaluated using different evaluation metrics, such as recall (Rel), accuracy (Acy), F-measure (F-me), and precision (Pre). In a binary classification problem, true positive (T_P) and true negative (T_N) denote malicious and normal samples are correctly distinguished. Similarly, false positive (F_P) denotes the error if a normal sample is identified as abnormal or malicious. False negative (F_N) denotes the number of abnormal samples identified as normal. These metrics are computed as follows:

$$\text{Acy} = \frac{T_p + T_n}{T_p + T_n + F_p + F_n} \tag{9}$$

$$\text{Pre} = \frac{T_p}{T_p + F_p} \tag{10}$$

$$\text{Rel} = \frac{T_p}{T_p + F_n} \tag{11}$$

$$F - \text{me} = 2 \times \frac{\text{Pre} \times \text{Rel}}{\text{Pre} + \text{Rel}}. \tag{12}$$

C. Results Discussion and Explanations

In this proposed framework, the two-step SW is utilized to preprocess source univariate ts. The first targets to convert 1-D ts into the 2-D data source, whereas the second SW can be reflected to mine temporal features at the low level that is used to fetch time dependencies to spatial attributes mined using the CL. Experiments are conducted to assess the influence of 2-step SW utilizing the combination of LSTM and CNN.

The methods used for evaluation and comparison purposes are: 1) simple CNN with no SW (N-CNN); 2) SW-based CNN (SW-CNN); 3) simple combination of CNN and LSTM with no SW (N-Conv-LSTM); and 4) SW-based combination of Conv-LSTM (SW-Conv-LSTM). All these methods are trained using training data for 150 epochs having a 0.01 learning rate. The comprehensive performances of these models are shown in Table III and Fig. 3. The trajectory of the loss values of the four methods in the learning process is shown in Fig. 3, where the blue line shows the loss value for the proposed model and the green, red, and cyan lines show the loss values for other models. Similarly, the evaluation scores of the tested methods are also plotted in Fig. 3 using the four different evaluation metrics.

TABLE III
PERFORMANCE COMPARISON OF PROPOSED AND OTHER METHODS

Method	Acy (%)	Pre (%)	Rel (%)	F-me (%)
CNN	94.32	94.89	94.32	94.33
LSTM	95.08	95.41	95.08	95.13
Conv-LSTM	98.26	98.29	97.17	98.21
Autoencoder-LSTM	98.76	99.17	97.82	99.14
Conv-LSTM-autoencoder	99.35	99.35	98.86	99.33

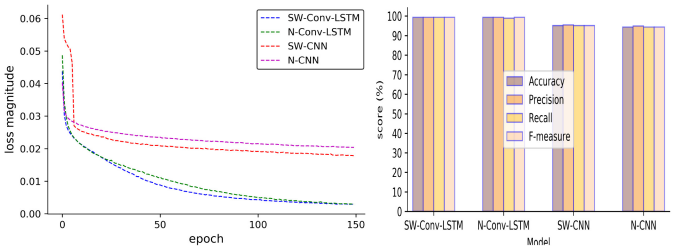


Fig. 3. Comparison of loss and evaluation metrics after the effect of two-step SW.

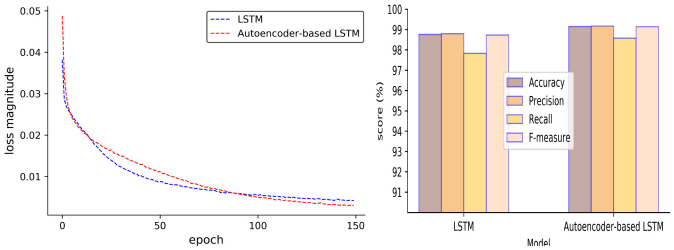


Fig. 4. Comparison of autoencoder-based LSTM and simple LSTM for loss and evaluation metrics.

We can observe from Fig. 3 and Table III that the combination of CNN and LSTM attains the lowest value for loss in the training process and achieved higher scores for accuracy, recall, precision, and f-measure. It can also be observed from the results that 2-step SW is inadequate for the simple CNN method, but the combined CNN and LSTM model produced better results. This may be because the mined features of CNN using one SW have no temporal usefulness, which makes it challenging for the LSTM model to extract temporal features. But, the two-step SW transforms every instance into several continuous fragments and delivers time dependencies for enhanced extraction of temporal features.

In [13], the C-LSTM method involves the combination of CNN and LSTM, where LSTM has been employed to mine temporal features at a high level. The proposed improved method switches the simple LSTM model by the autoencoder-based LSTM model to accomplish enhanced extraction of temporal features. As proof of concept, we conducted experiments and compared the results of both the simple LSTM method and autoencoder-based LSTM model. These two methods are trained using training data sources comprising both normal and abnormal data for 150 epochs having a 0.001 rate of learning. The comparative results are shown in Fig. 4, where it can be observed that the autoencoder-based LSTM model achieved better performance in terms of all evaluation metrics and reached the lowest value of the loss. Thus, approving that the autoencoder-based LSTM model could mine enhanced features for better classification results.

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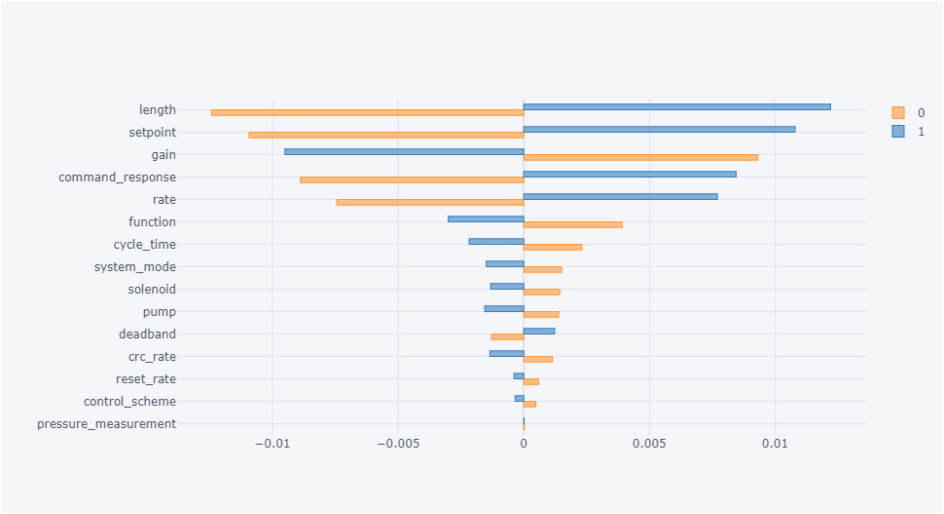


Fig. 5. Inference graph across the two classes. This visualization shows which feature was more vital in predicting a specific class.

TABLE IV
PERFORMANCE COMPARISON USING RECONSTRUCTION ERROR

Method	Acy (%)	Pre (%)	Rel (%)	F-me (%)
KNN	92.63	95.67	66.80	80.09
RF	86.66	92.68	39.89	57.03
SVM	89.21	93.87	51.43	67.91

Malhotra *et al.* [36] trained autoencoder-based LSTM using only the normal data set and recognized abnormality detection process using the original and decoded data error rates. They presume that the error vectors have a normal distribution and consume estimation of maximum likelihood for parameters estimations. Nevertheless, we discovered that this technique is inappropriate for GSP system data since there is only a trivial variation between regular and malicious data. Also, because only normal instances are used in the training process, the reconstruction error is not sufficient to detect malicious instances correctly. To validate this idea, the autoencoder-based LSTM is trained using normal data only in the training process, and reconstruction errors are acquired using testing samples covering both of the two data categories.

The reconstruction error of both normal data and anomalous data are computed by training the autoencoder-based LSTM model. The reconstruction error of the decoded instances appears similar to that of the original records, which occurs in minor errors. To assess the impact of classification results using error of reconstruction, we used several ML algorithms, such as KNN, RF, and SVM to categorize using error vectors. These models were obtained from the SciKit-Learn repository, which offers simple and proficient employment of ML methods. These methods are directly applied using the conventional parameter set in the SciKit-learn library.

The outcomes of the classification using error vectors are outlined in Table IV, where it can be observed that all the compared methods achieved a high accuracy rate but a low score for F-measure affected by imbalance categories. The Low values of recall and F-measure indicate that it is difficult to identify malicious data using reconstruction errors. Consequently, we have chosen to train the autoencoder-based LSTM model using normal and anomalous records and reflect it as feature mining at high-level for improved classification, instead of increasing the dissimilarity between two data categories (normal and anomalous).

From the analysis of the evaluation results discussed above, we put forward the enhanced anomaly discovery method for ts using the combination of CNN, autoencoder-based LSTM model, and DNN. Before ts is fed into the proposed enhanced method, it is preprocessed by utilizing a two-step SW. Comparing several anomaly detection methods using experiments demonstrates the proposed framework's dominance in terms of all performance metrics. The explanations of detections are shown in Figs. 5 and 6. Fig. 6 aggregates the overall effect of features and gives the idea of what features are significantly relevant and important in the larger view. Similarly, Fig. 5 separates the inference across the two classes and looks at them distinctly. This Fig. 5 presents which feature was more imperative in forecasting a certain class.

Furthermore, the processing of SW can be reflected as extracting information about low-level temporal features, and the size (S) of SW is the key for the subsequent convolutional process and classification results. The detection performance using diverse sizes of SW is shown in Fig. 7, where the proposed framework achieved the best results using the size of SW as $S = 8$. It can also be observed from Fig. 7 that the detection performances are not good with too small or too large size of SW. The large size SW means that SW has not treated it at the second step, which could miss the detailed information, and the abnormalities could be comparatively regular in a smaller size SW. Hence, picking a suitable window size is a challenging and significant task in TSDM.

D. Implications and Applicability

This study delivers valuable practical implications for adopting explainable intelligent anomaly detection-based security solutions to protect IIoT-based industrial networks. The developed detection framework can assist as an adequate start to build vigorous and trustworthy anomaly detection models

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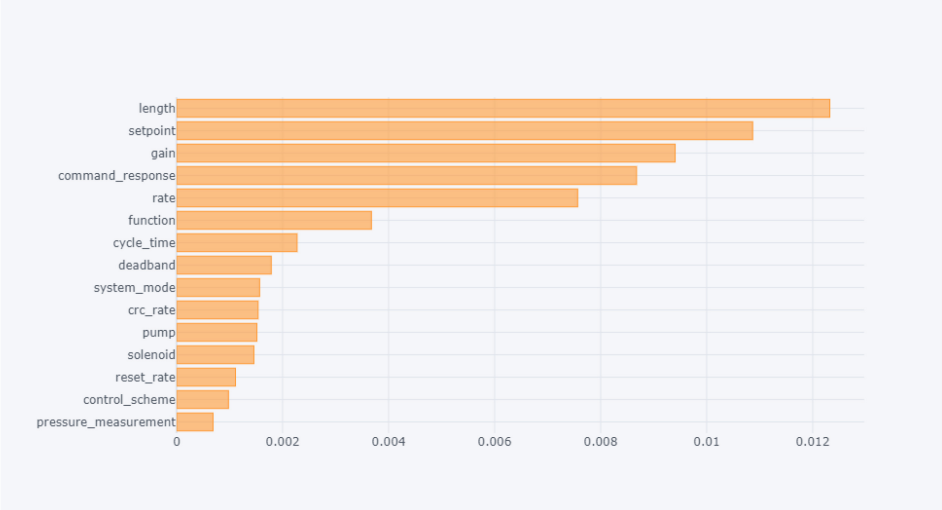


Fig. 6. Global explanation of detection framework. It displays feature ranking ordered from uppermost to lowermost. The magnitude of ranking is shown in the x axis.

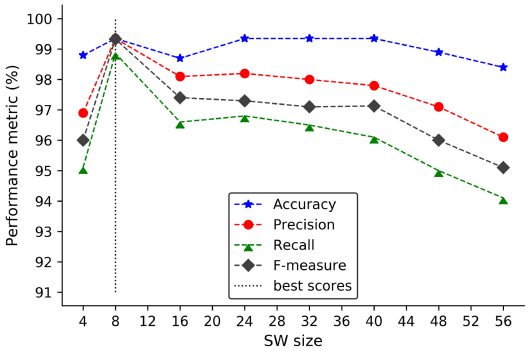


Fig. 7. Performance comparison of proposed framework using different S values of SW.

instead of traditional black-box models. The assessment results offered in Section IV-C have provided insights into recurrent convolutional autoencoders' effectiveness.

The analysis done in this work can support the speedy evolution of shifting traditional nonexplainable DL methods to explainable ones. This is because knowledge of anomalies and susceptibility to discovery is vital for the success of forthcoming detection systems. Previous studies in anomaly detection and IIoT networks domain just focused on detection algorithms to express the usefulness of their approach. However, our proposed model is different because it provides a way to understand the prediction decisions. Although the detection of attack is imperative, reasoning and understanding how they are prompted is an imperious element. Any security administrator who tries to investigate the attacks has a "right to a clarification and interpretation of the detection decision." This research work delivers this by imitating human-level rationale to simplify security administrators' interpretation of the prediction decisions, causal reasoning, and underlying data evidence.

The novel contributions of this proposed framework also lie in the implications and applicability of the proposed model in industrial aspects, such as Industry 4.0 and IIoT-based networks. In contrast, Kim and Cho's [13] work is mainly focused on Web traffic detection. To the best of our knowledge, at the time of writing, there is no published work that detects anomalies and explain the prediction results in Industry 4.0 or IIoT domain. The proposed method can be applied to diverse industrial systems, such as power systems, water storage/treatment systems, etc., using an offline training process and then detecting anomalies in real time. The dynamic nature of the proposed method can be enhanced and fine-tuned to be applied in different ICSs.

V. Conclusion and Future Work

This study has proposed a novel framework using the integration of CNN and autoencoder-based LSTM method to accomplish improved detection of anomalies in IIoT-based networks. We have demonstrated that the two-step SW process can extract better temporal features using LSTM, which collects the CNNs' output in the integrated framework. This two-step SW phenomenon can be reflected as extracting features at a basic temporal level. CNN is employed to extract spatial features at a high level, and LSTM is utilized to extract temporal features at a high level. The proposed detection framework applied an autoencoder-based LSTM model to mine better features to improve classification results in a fully connected way. The experimental results on real GSP data validated that the proposed detection framework accomplished a high accuracy rate, recall, precision, and F-measure in detecting anomalies in IIoT-based networks. Moreover, we employed explainable AI techniques to improve trust management that security administrators can interpret (such as causal reasoning and underlying data evidence). The model needs to be tested on other large-scale industrial data sets (such as power systems) to test its efficiency and generalization. In the future, we plan to enhance this model using attention architecture and resolve the issue of detecting anomalies in multivariate time series.

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